EDS 223 HW 3: Prioritizing potential aquaculture

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```
library(terra)
library(tidyverse)
library(sf)
library(tmap)
library(kableExtra)
library(spData)
library(spDataLarge)
library(geodata)
```

```
# Read in the data
wc <- st_read('data/wc_regions_clean.shp')
```

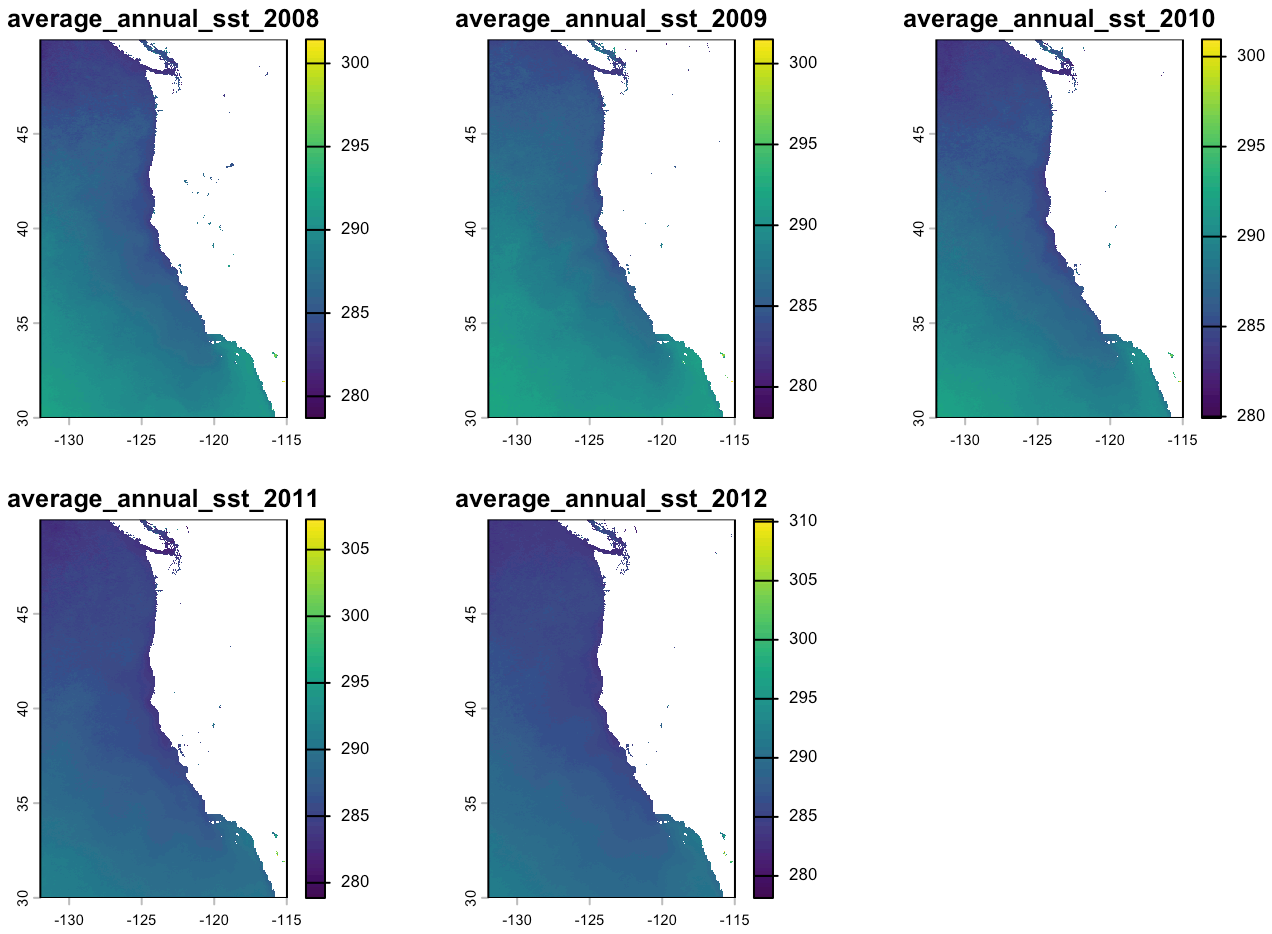
```
Reading layer `wc_regions_clean' from data source
  `/Users/emaliapartlow/Desktop/MEDS/eds223/Prioritizing-potential-
aquaculture/data/wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```
sst_2008 <- rast('data/average_annual_sst_2008.tif')
sst_2009 <- rast('data/average_annual_sst_2009.tif')
sst_2010 <- rast('data/average_annual_sst_2010.tif')
sst_2011 <- rast('data/average_annual_sst_2011.tif')
sst_2012 <- rast('data/average_annual_sst_2012.tif')
depth <- rast('data/depth.tif')
```

```
# Compare SpatRaster geometries
compareGeom(
  sst_2008,
  sst_2009,
  sst_2010,
  sst_2011,
  sst_2012,
  ext = TRUE,
  res = TRUE,
  crs = TRUE
)
```

```
[1] TRUE
```

```
# Stack the rasters
sst <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)
plot(sst) # Check stack worked
```




```
# Check if all dataframe's CRS match
if ((st_crs(wc) == crs(sst)) & (crs(sst) == crs(depth))) {
  print("It's a match")
} else {
  warning("Coordinate refrence systems do not match")
}
```

```
# Reproject depth to match sst CRS
depth_reprojected <- project(depth, sst)

# Reproject wc to match sst CRS
wc_reprojected <- st_transform(wc, crs(sst))
```

```
# Double check all dataframe's CRS match
if ((crs(wc_reprojected) == crs(sst)) & (crs(sst) == crs(depth_reprojected))) {
  print("It's a match")
} else {
```

```
  warning("Coordinate refrence systems still do not match")
}
```

```
[1] "It's a match"
```

```
# Take the mean SST from 2008-2012
mean_sst <- app(sst, fun = mean, na.rm = TRUE)

# Convert average SST from Kelvin to Celsius
mean_sst_c <- mean_sst - 273.15
```

Note: Did not need to manually crop or resample because project() did that

```
# Double check that everything matches
cat("Resolutions match:", all(res(mean_sst_c) == res(depth_reprojected)), "\n")
```

```
Resolutions match: TRUE
```

```
cat(
  "Origins match:    ",
  all(round(origin(mean_sst_c), 8) == round(origin(depth_reprojected), 8)),
  "\n"
)
```

```
Origins match:     TRUE
```

```
cat("Extents match:    ", ext(mean_sst_c) == ext(depth_reprojected), "\n")
```

```
Extents match:     TRUE
```

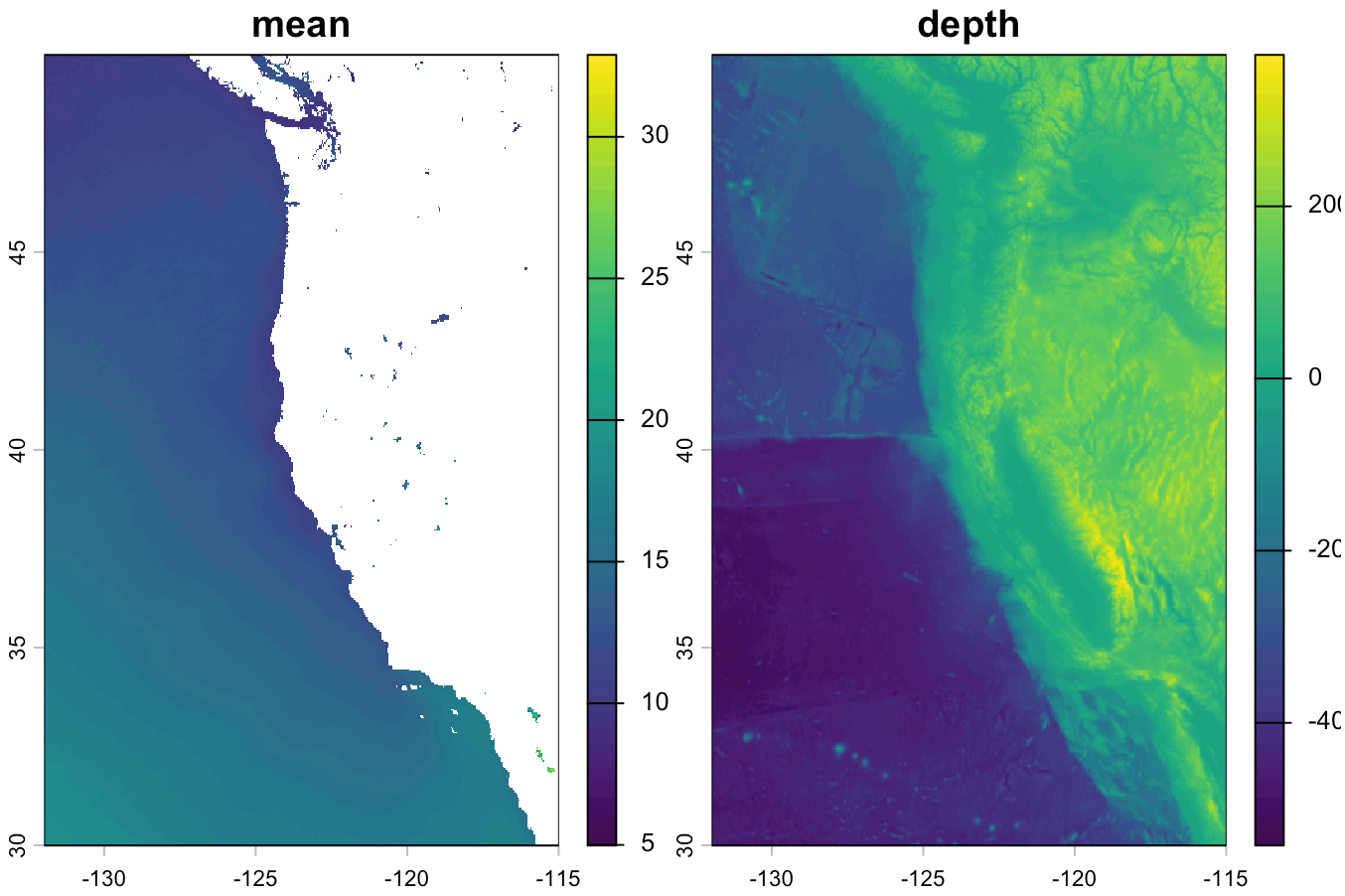
```
cat("CRS match:        ", crs(mean_sst_c) == crs(depth_reprojected), "\n")
```

```
CRS match:         TRUE
```

```
cat("Dimensions match:  ", all(dim(mean_sst_c) == dim(depth_reprojected)), "\n")
```

```
Dimensions match:   TRUE
```

```
# Stack rasters
sst_depth <- c(mean_sst_c, depth_reprojected)
plot(sst_depth)
```



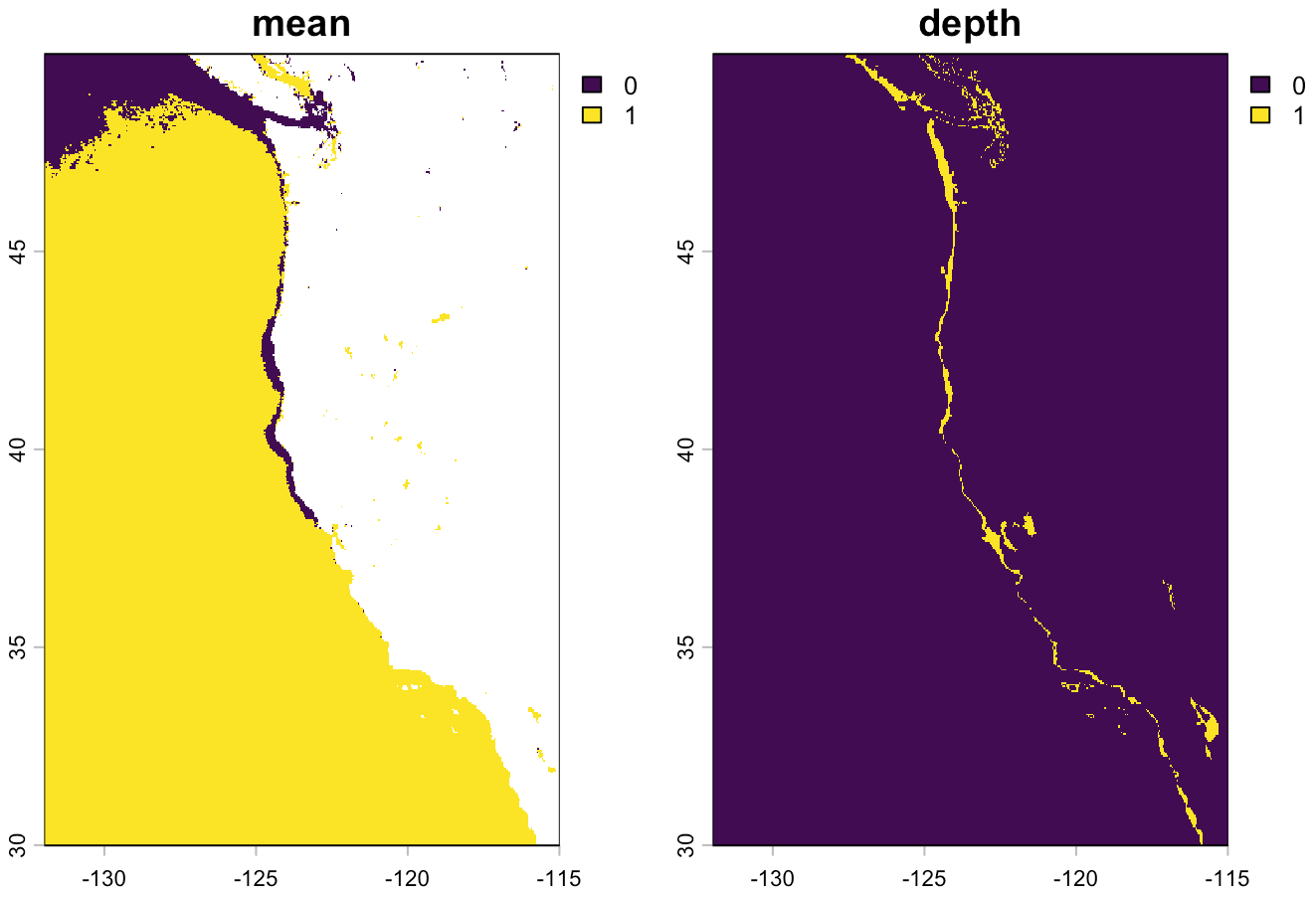

```r
# Reclassify SST and depth data into locations that are suitable for oysters
rcl_depth <- matrix(
   c(
      -Inf,
      -70,
      0, # Not suitable
      -70,
      0,
      1, # Suitable
      0,
      Inf,
      0 # Not suitable
   ),
   ncol = 3,
   byrow = TRUE
)

rcl_sst <- matrix(
   c(
      -Inf,
      11,
      0, # Not suitable
      11,
```

```
      30,
      1, # Suitable
      30,
      Inf,
      0 # Not suitable
   ),
   ncol = 3,
   byrow = TRUE
)

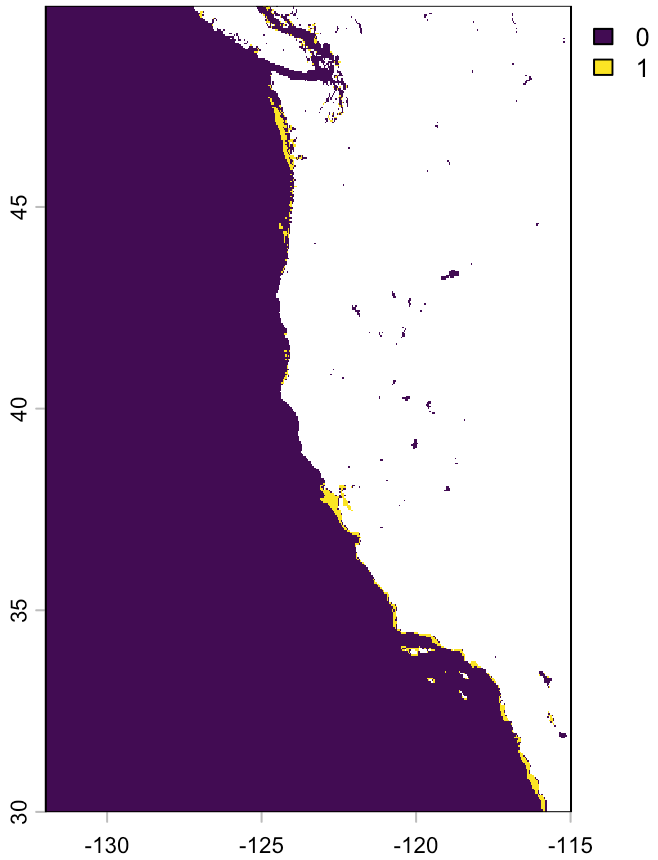
depth_zones <- classify(depth_reprojected, rcl = rcl_depth)
sst_zones <- classify(mean_sst_c, rcl = rcl_sst)

# Stack rasters
suitable_zones <- c(sst_zones, depth_zones)
plot(suitable_zones)
```




```
# Locations that satisfy both SST and depth conditions
best_zone <- lapp(suitable_zones, fun = function(s, d) s * d)
names(best_zone) <- "suitable_sst_depth"

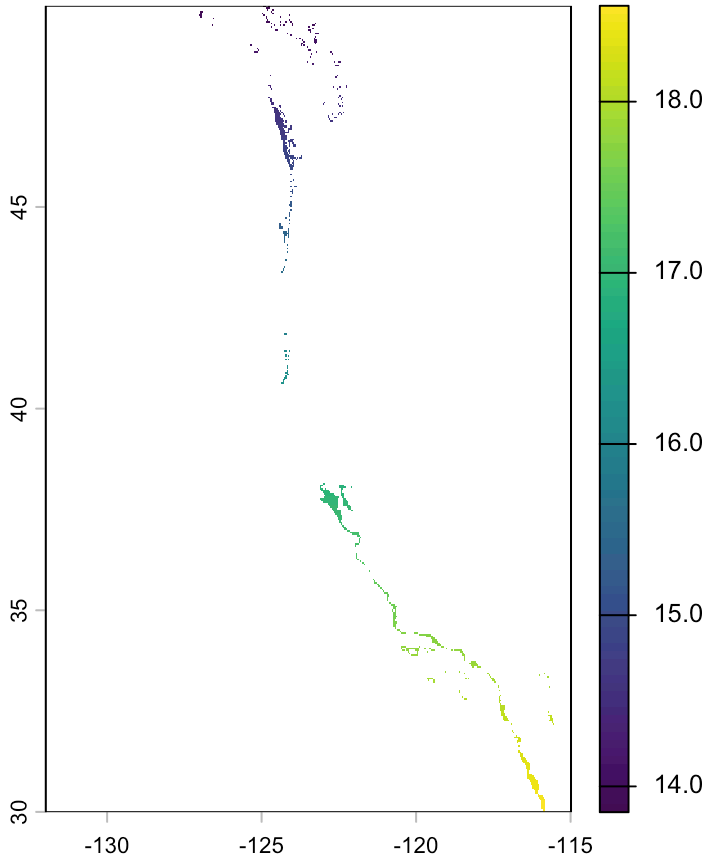
plot(best_zone)
```




```
# Convert EEZ regions to raster cells
eez_raster <- rasterize(wc_reprojected, best_zone, field = 'rgn_id')

# Find area of cells
area <- cellSize(best_zone, unit = 'km', mask = TRUE)

# Select suitable depth and sst zone cells within West Coast EEZs
eez_masked <- mask(area, best_zone, maskvalue = 0) # Convert 0 values to NA
plot(eez_masked)
```




```
# Summarize cell size variation across the extent
zonal_area <- zonal(eez_masked, eez_raster, fun = "sum", na.rm = TRUE)
```

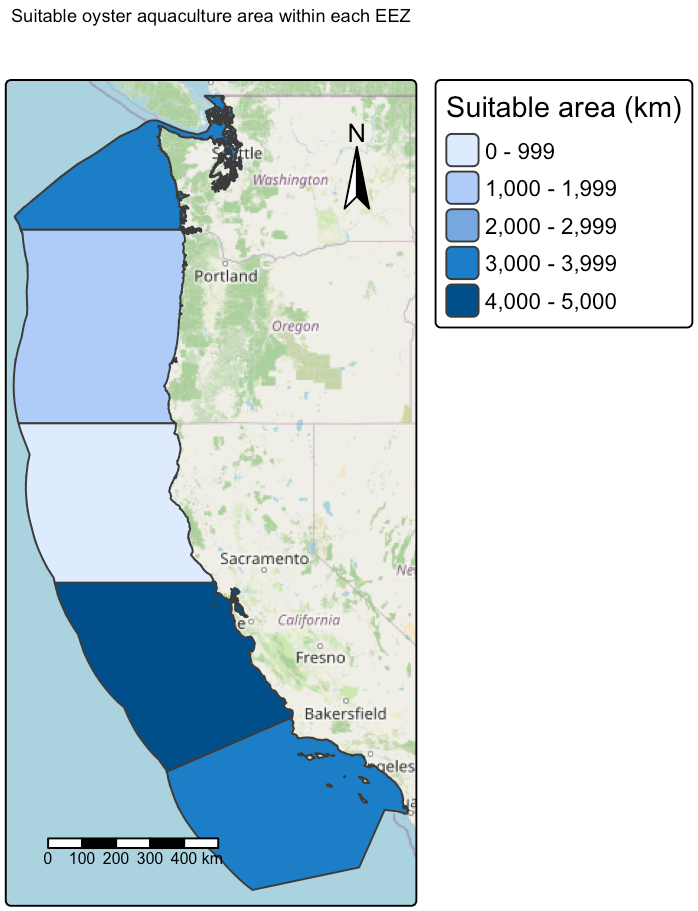
```
kable(zonal_area, col.names = c("EEZ", "Suitable area (km)"), digits = 1) |>
  kable_styling(bootstrap_options = "striped")
```

Suitable area for marine aquaculture within each West Coast EEZ

EEZ	Suitable area (km)
1	1395.7
2	323.9
3	4373.6
4	3329.9
5	3091.5

```
# Join suitable area onto original vector eez data frame
map <- left_join(wc_reprojected, zonal_area, by = 'rgn_id')
```

```
# Create a map of EEZ regions colored by amount of suitable area
tm_shape(map) +
  tm_basemap('OpenStreetMap') +
  tm_polygons(
    fill = 'area', # Fill polygons based on suitable area
    fill.legend = tm_legend(
      title = 'Suitable area (km)',
      position = tm_pos_out('right', 'center')
    )
  ) +
  tm_compass(position = c('right', 'top')) +
  tm_scalebar(position = c('left', 'bottom')) +
  tm_title('Suitable oyster aquaculture area within each EEZ') +
  tm_legend(position = tm_pos_out('right', 'center'))
```



The oyster map displays that the central California EEZ has the highest amount of suitable area for oyster aquacultre. The northern California EEZ has the least amount of suitable area.

```
# Create a function of the workflow that can be applied to different species
suitable_aquaculture_eez <- function(
  species,
  sst_min,
  sst_max,
```

```
  depth_min,
  depth_max
) {
  # Stack sst rasters
  sst <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012)

  # Reproject depth to match sst CRS
  depth_reprojected <- project(depth, sst)

  # Reproject wc to match sst CRS
  wc_reprojected <- st_transform(wc, crs(sst))

  # Take the mean SST from 2008-2012
  mean_sst <- app(sst, fun = mean, na.rm = TRUE)

  # Convert average SST from Kelvin to Celsius
  mean_sst_c <- mean_sst - 273.15

  # Stack depth and sst rasters
  sst_depth <- c(mean_sst_c, depth_reprojected)

  # Reclassify SST and depth data into locations that are suitable for the species
  rcl_depth <- matrix(
    c(
      -Inf,
      depth_min,
      0, # Not suitable
      depth_min,
      depth_max,
      1, # Suitable
      depth_max,
      Inf,
      0 # Not suitable
    ),
    ncol = 3,
    byrow = TRUE
  )

  rcl_sst <- matrix(
    c(
      -Inf,
      sst_min,
      0, # Not suitable
      sst_min,
      sst_max,
      1, # Suitable
      sst_max,
      Inf,
      0 # Not suitable
    ),
    ncol = 3,
```

```
    byrow = TRUE
  )

  # Apply zone classification
  depth_zones <- classify(depth_reprojected, rcl = rcl_depth)
  sst_zones <- classify(mean_sst_c, rcl = rcl_sst)

  # Stack classified rasters
  suitable_zones <- c(sst_zones, depth_zones)

  # Calculate locations that satisfy both SST and depth conditions
  best_zone <- lapp(suitable_zones, fun = function(s, d) s * d)
  names(best_zone) <- "suitable_sst_depth"

  # Convert EEZ regions to raster cells
  eez_raster <- rasterize(wc_reprojected, best_zone, field = 'rgn_id')

  # Find area of cells
  area <- cellSize(best_zone, unit = 'km', mask = TRUE)

  # Select suitable depth and sst zone cells within West Coast EEZs
  eez_masked <- mask(area, best_zone, maskvalue = 0)

  # Summarize cell size variation across the extent
  zonal_area <- zonal(eez_masked, eez_raster, fun = "sum", na.rm = TRUE)

  # Join suitable area onto original vector eez data frame
  map <- left_join(wc_reprojected, zonal_area, by = 'rgn_id')

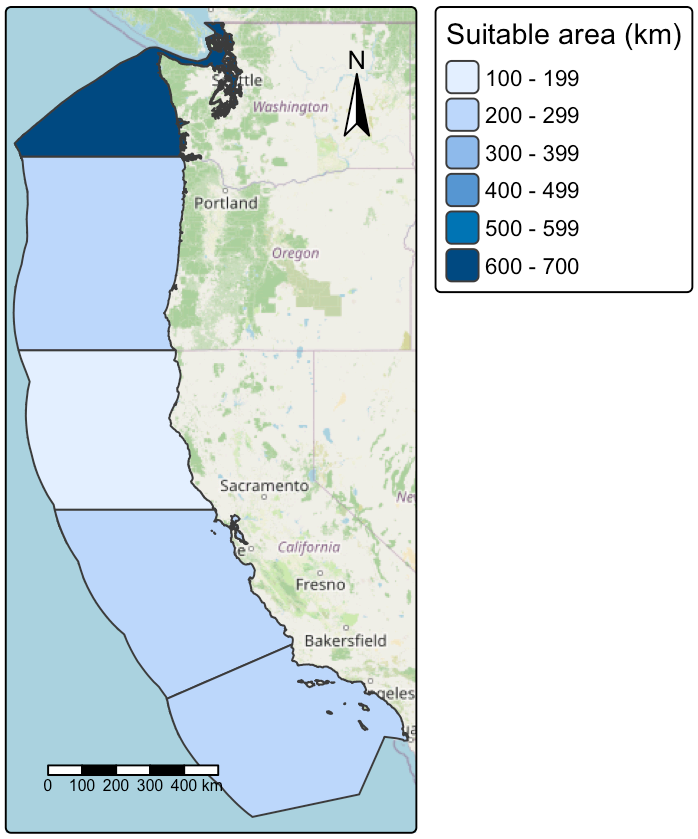
  # Create a map of EEZ regions colored by amount of suitable area
  aquaculture_map <- tm_shape(map) +
    tm_basemap('OpenStreetMap') +
    tm_polygons(
      fill = 'area', # Fill polygons based on suitable area
      fill.legend = tm_legend(
        title = 'Suitable area (km)',
        position = tm_pos_out('right', 'center')
      )
    ) +
    tm_compass(position = c('right', 'top')) +
    tm_scalebar(position = c('left', 'bottom')) +
    tm_title(paste('Suitable', species, 'aquaculture area within each EEZ')) +
    tm_legend(position = tm_pos_out('right', 'center'))

  return(aquaculture_map)
}
```

```
suitable_aquaculture_eez(
  species = 'red abalone',
  sst_min = 10,
```

```
  sst_max = 18,
  depth_min = 6,
  depth_max = 40
)
```

Suitable red abalone aquaculture area within each EEZ



The red abalone map displays that the Washington EEZ has the highest amount of suitable aquaculture area. The northern California EEZ has the least amount of suitable area.